

Introduction to Pure Mathematics

Problem Sheet 1

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Section B (7)

Let $a, b \in \mathbb{R}$ with $a < b$. Show that $\frac{a+b}{2} < b$ and $a < \frac{a+b}{2}$. Arithmetic properties of $\mathbb{R}$ these are satisfied $\forall x, y, z \in \mathbb{R}$

$$(\mathbb{R}1) \ x \cdot 0 = 0, 1 \cdot x = x \text{ and } x + 0 = x$$

$$(\mathbb{R}2) \ -(-x) = x \text{ and } -0 = 0$$

$$(\mathbb{R}3) \ y + x = x + y \text{ and } x + (y + z) = (x + y) + z$$

$$(\mathbb{R}4) \ \text{if } x > y \text{ then } x + y > y + z; \text{ in addition, if } z > 0 \text{ then } xz > yz$$

$$(\mathbb{R}5) \ \text{if } x > y \text{ and } y > z \text{ then } x > z$$

$$(\mathbb{R}6) \ x^2 \geq 0$$

Proof: Starting with $a < b$ we can add b to both sides due to $\mathbb{R}4$ giving us $a + b < 2b$ then multiply both sides by $\frac{1}{2}$ since $\frac{1}{2} > 0$ giving us $\frac{a+b}{2} < b$. In order to prove the second statement we again start with $a < b$ then add a using $\mathbb{R}4$ giving $2a < a + b$ and again times through by $\frac{1}{2}$ giving $a < \frac{a+b}{2}$. Therefore $\forall a, b \in \mathbb{R}$ if $a < b$ then $\frac{a+b}{2} < b$ and $a < \frac{a+b}{2}$.